What a grid slot is worth

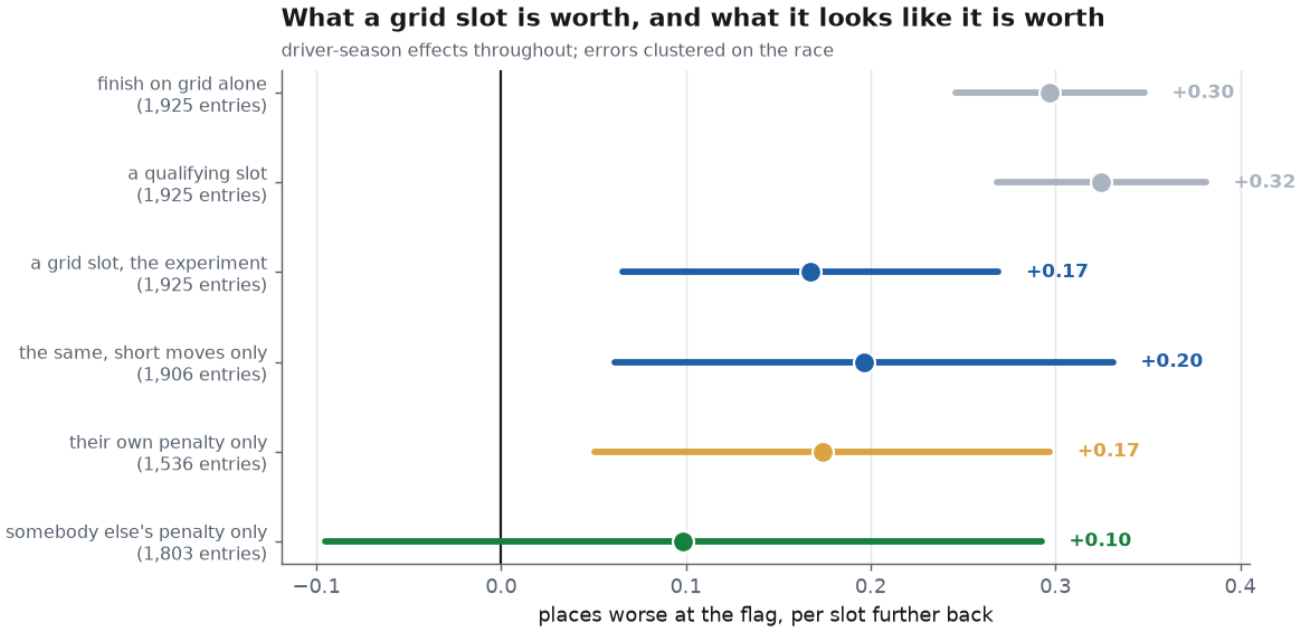
Formula 1, 2018–2023. Measured from grid penalties — the one thing in the sport that moves a car without changing the car, the driver or the weekend.

A grid slot is worth +0.17 places at the flag

95% interval [+0.066, +0.269]. Regressing on the grid alone says +0.297, and that is already inside a driver's own season — so a large part of what looks like track position is the car reasserting itself over the race distance.

+0.17	+0.30	596	146
a grid slot is worth	it looks worth	entries moved by a penalty	points it cost them

Where a car starts predicts where it finishes almost perfectly, and almost none of that is the starting place: quick cars start at the front. Keep both numbers — where the car earned its slot in qualifying and where it actually started — and the difference between them is a shift the car's own pace did not cause.



The experiment

Within a driver's own season and at a given qualifying position, the only thing moving that shift is a steward or somebody else's failed gearbox.

The pace premium — what a qualifying slot is worth beyond the track position it comes with — is +0.157 places a slot, 95% [+0.041, +0.274], p = 0.009.

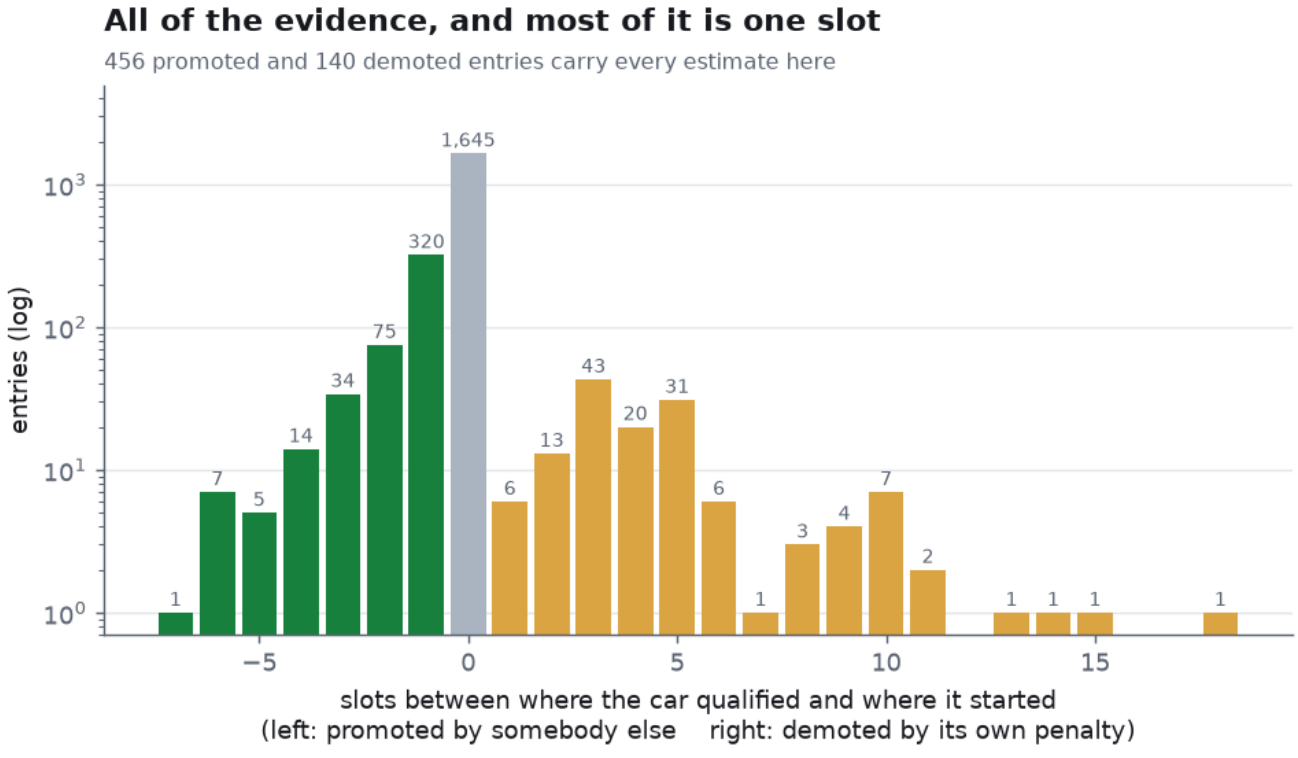
what is being measured	entries	places a slot	95% interval	excludes zero
finishing position on grid alone	1,925	+0.297	[+0.246, +0.348]	yes
a qualifying slot	1,925	+0.325	[+0.268, +0.381]	yes
a grid slot, the experiment	1,925	+0.167	[+0.066, +0.269]	yes
the same, short moves only	1,906	+0.196	[+0.061, +0.331]	yes
their own penalty only	1,536	+0.174	[+0.051, +0.297]	yes
somebody else's penalty only	1,803	+0.099	[-0.095, +0.292]	no

All of the evidence

And the honest limit of the design.

A promotion is nearly always a single slot, so the estimate fitted on promoted cars alone — the only cars moved for a reason that has nothing to do with them — has an interval wide enough to contain both zero and the pooled answer. It is not a refutation and it is not a confirmation.

Fitting both directions in one design makes the comparison testable: backwards +0.183, forwards -0.117. If the effect were symmetric these would sum to zero; they sum to +0.066 ± 0.119, p = 0.581 — consistent with symmetry, so the thin promoted estimate is not evidence against the demoted one.

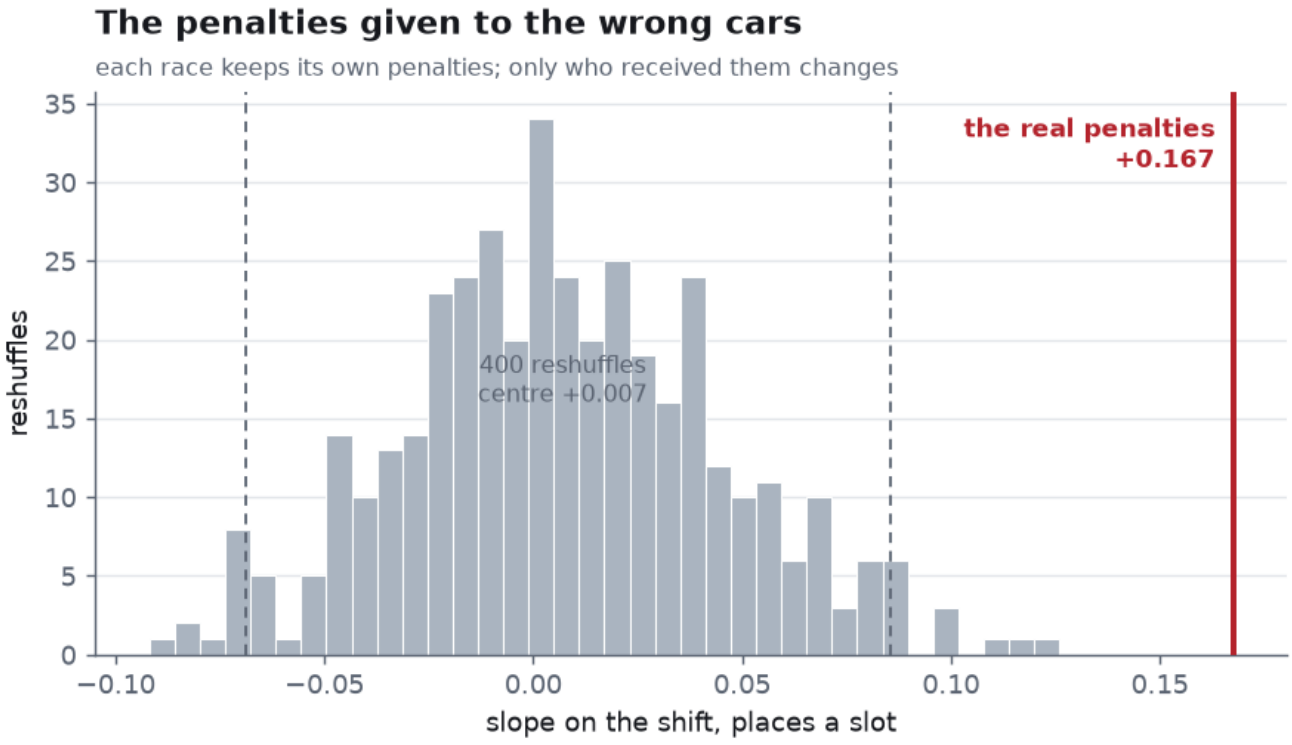


Two ways to catch this inventing an effect

Within each race, give the penalties to the wrong cars. Every race keeps exactly the penalties it really had; only who received them is destroyed.

0 of 400 reshuffles reached the real slope, p = 0.0025, and they centre on +0.007.

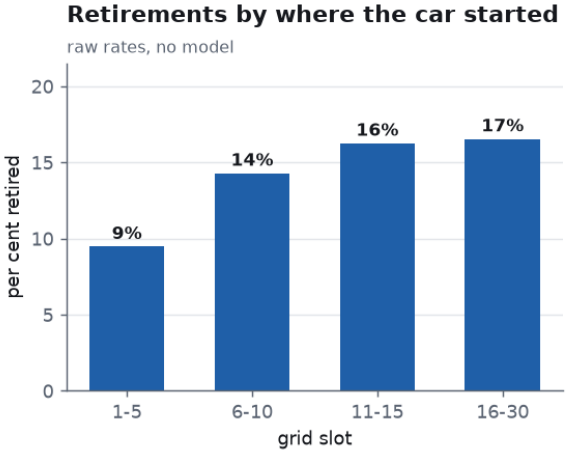
Then the same fit, predicting a race that already happened. A penalty handed out on Saturday cannot change what the same driver did a fortnight earlier, so the slope on the driver's previous finishing position has to be zero: it is -0.027 ± 0.048, p = 0.581. Anything else would mean penalties go to drivers whose results were already moving.

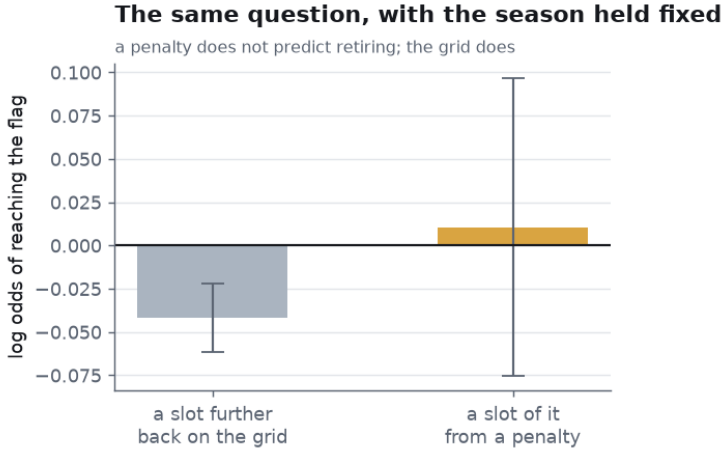


The exclusion that could bend it

A retirement has no finishing position, so 316 of 2,241 entries are not in the main fit.

A grid slot does predict retiring (odds ratio 0.959 a slot, p = 0.000), but a penalty does not (odds ratio 1.011, p = 0.809). The first is a selection effect on the outcome and makes the main slope an understatement; the second is the exogeneity the design needs, and it holds.





grid slot	entries	retired	share
1-5 (front five)	570	54	9.5%
6-10 (sixth to tenth)	567	81	14.3%
11-15 (eleventh to fifteenth)	566	92	16.3%
16-30 (sixteenth and back)	538	89	16.5%

The logit carries season effects rather than driver-season effects because many driver-seasons contain no retirement at all, which predicts the outcome perfectly and leaves the likelihood with no finite maximum.

The held-out set that turned out not to exist

Pit-lane starts are excluded from every fit, because the pit lane is not a grid slot. The intention was to use them as a held-out set, moved further than any penalty in the panel.

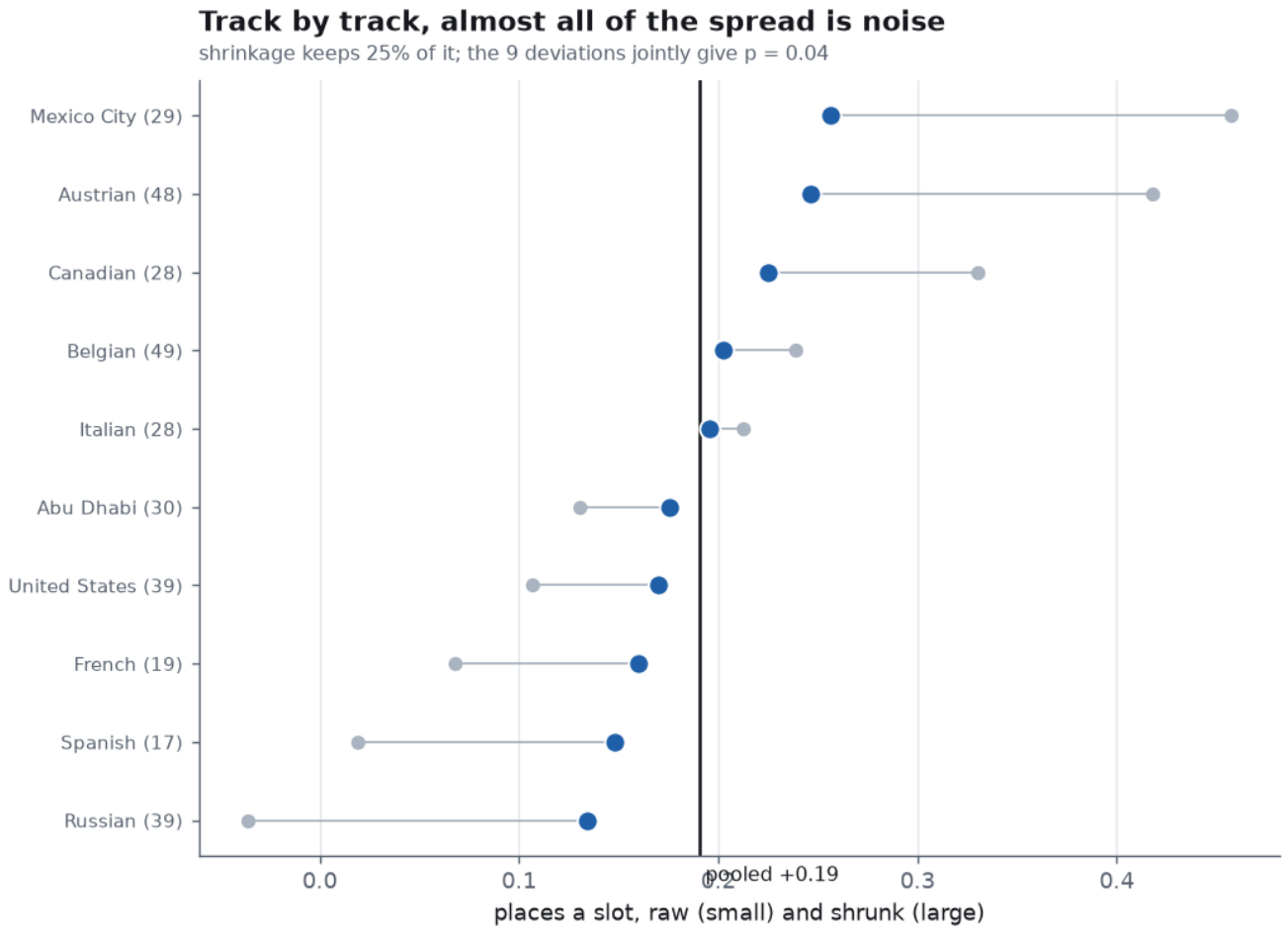
Almost none of them were moved at all. Their median distance from where they qualified is 2 slots, and only 12 of 31 classified starts were moved 5 or more, because the failure that earns a pit-lane start is usually the failure that ruined qualifying: these cars had already qualified at the back. For a few the distance is negative — when several cars start from the pit lane the grid is shorter than the qualifying order, so a car that qualified last was going to start last regardless.

A pit-lane start is a severe punishment in every respect except the one measured here. Over the moved ones the implied slope is +0.335. 5 of 36 of these starts retired, against 14% in the panel.



Track by track

Shrinkage keeps 25% of the spread between circuits and attributes the rest to sampling noise. Jointly the 9 deviations give F = 2.29, p = 0.035.



circuit	entries	moved	slope	error	shrunk
Mexico City	87	29	+0.457	0.113	+0.256
Austrian	100	48	+0.418	0.164	+0.247
Canadian	69	28	+0.330	0.187	+0.225
Belgian	83	49	+0.239	0.112	+0.202
Italian	82	28	+0.212	0.064	+0.196
Abu Dhabi	107	30	+0.131	0.078	+0.176
United States	64	39	+0.107	0.070	+0.170
French	72	19	+0.068	0.068	+0.160
Spanish (baseline)	107	17	+0.019	0.275	+0.148
Russian	69	39	-0.036	0.167	+0.134

Where the penalties happen is a choice

A team that must change a power unit decides when, and it picks a weekend where starting further back is cheap.

So the variation this entire project is identified from is not spread evenly over the calendar. Every number here answers "what was a slot worth where penalties were actually taken", which is the right question for pricing a penalty and a narrower one than pricing a grid slot.

> The weighting is not where this bites: pooled over the eligible circuits the slope is +0.190 and the unweighted mean of their own slopes is +0.195, so the busy circuits are not dragging the pooled number about. What bites is the 21 circuits that cannot be fitted at all — Monaco among them, where a slot should be worth most of anywhere. They are missing rather than down-weighted.

circuit	entries	moved	share moved
Las Vegas	20	17	85%
Turkish	38	25	66%
United States	78	46	59%
Belgian	95	55	58%
. . .			
British	119	12	10%
Hungarian	116	8	7%
Japanese	79	4	5%
Portuguese	40	0	0%

What it cost, and what this does not do

116 demoted entries reached the flag inside the 10-slot support, 489 slots between them, with 6 moved further and refused rather than extrapolated, for 146.0 championship points — 1.26 an entry, with only 56 of them losing anything at all, because most penalties land outside the points.

> This is what the penalised cars gave up, not what the grid rearranged. Un-penalising one car moves every car it passes, and those counterfactuals are not computed, so the gains on the other side of these losses are missing.

It is a local, linear average over the shifts that happened, nearly all of them between one and ten slots. It cannot say why a car was moved, because the feed records no reason: gearbox, power unit, impeding and blocking are one column. Sprint weekends are dropped entirely. And the promoted-only estimate, the one with no contamination in it, is too imprecise to settle the question on its own.

The panel

Qualifying, grid and race results from the Jolpica mirror of the Ergast schema.

	entries
entries in the feed	2,500
dropped, sprint weekend	220
dropped, no qualifying record	3
set aside, started from the pit lane	36
in the panel	2,241
of those, classified at the flag	1,925
of those, moved by a penalty	596